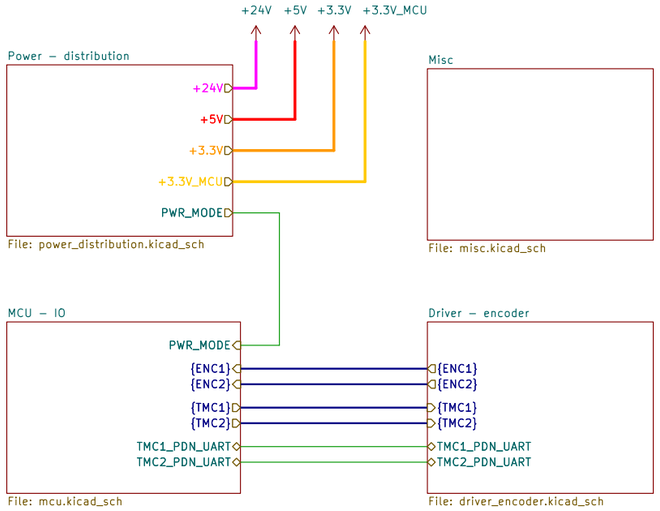
Dual Stepper Driver Evaluation Board

2026-10-22
Rev 1.0



To-Do:

During PCB design phase:
- simulations

Ordering:
- Check JLCPCB preview PCB

Final design:
- Implement impedance control USB
- should split IO in seperate sheet
- IPC design rules
- extensive pcb simulation in Altium
- block diagram
- firmware info? State diagram?

#	Test	Tools (IRL, Dev-Board, SIM, CALC)	Procedure	Pass criterion	Result
1	24V supply	Calculation + IRL	24 V only, no motors	+5 V within 4.85–5.15 V, +3.3 V within 3.20–3.40 V, ripple below 50 mVpp	
2	USB supply	IRL	USB only, with a C–to–C cable and an A–to–C cable	Board is recognised over USB with both; drivers unpowered	
3	Supply switching	Simulation + IRL	Unplug/replug 24 V while USB is connected, then unplug USB while 24 V is connected; scope +3.3V_MCU	No MCU reset, no USB disconnect, +3.3V_MCU stays above 3.0 V	
4	Reverse polarity	Simulation + IRL	–24 V from a lab supply limited to 0.5 A, for 10 s	Draws under 1 mA; board works afterwards	
5	Hot–plug	Simulation + IRL	Plug in the adapter with the 4 m cable 10 times; scope TP48	Peak below 29 V	
6	Brown–out	(Simulation +)IRL	Ramp a lab supply 0 → 24 → 0 V slowly, motors connected	No motor movement; clean start and stop	
7	Motor enable	Dev Board + IRL	Power up, hold reset, flash firmware, all with 24 V present	No holding torque until the firmware enables the motors	
8	Driver UART	Dev Board + IRL	Write and read back GCONF and IHOLD_IRUN on both drivers	Both answer, values match, no error flags	
9	Motor current	Calculation + IRL	Maximum set current; measure across the sense resistor (I = V / 0.22 Ω)	Within ±10 % of the setting and at most 1.0 A RMS	
10	Temperature	Calculation + IRL	Both motors at maximum load for 30 min, board in its enclosure	No overtemperature warning, TMC2209 below 90 °C, other parts below 70 °C	
11	Braking energy	Calculation + Simulation + IRL	Hardest deceleration at maximum load, and cut 24 V while running	+24 V stays below 29 V	
12	Encoder count	Dev Board + IRL	100 revolutions forward and back at maximum speed	Back to exactly 0 counts; one index pulse per revolution; direction correct	
13	Lockstep	Dev Board + IRL	Both motors driving the ring gear	Difference between the two encoders below 20 counts	
14	Encoder fault	Simulation + IRL	Unplug an encoder while running	No false counts; firmware flags the fault	
15	USB under load	IRL	Stream data for 30 min during maximum acceleration	0 errors, 0 disconnects	
16	Programming	IRL	Flash over SWD with the ST–LINK; enter DFU with BOOT0	Both work	
17	ESD	IRL	±4 kV contact discharge on the USB shell and the terminals	Keeps working or recovers by itself	

Standard passive component values

If not mentioned otherwise
R: tolerance=1%, package=0603
C: dielectric (not polarized)=X7R, package=0603

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for informational design notes.

DESIGN NOTE:
Example text for debug design notes.

ELECTRICAL CHARACTERISTICS:
Voltage supplied, Current drawn, ...

DESIGN NOTE:
Example text for critical design notes.

LAYOUT NOTE:
Example text for critical layout guidelines.

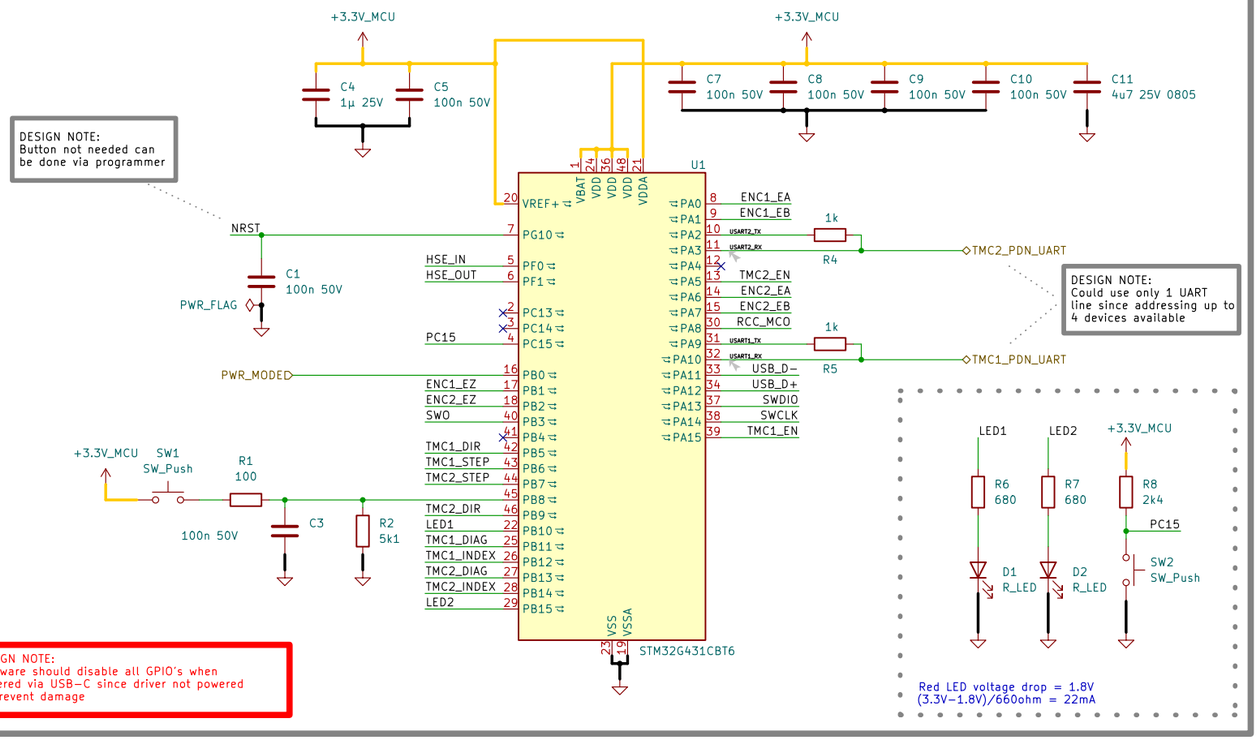
Sheet: /
File: stepper_motor_eval_board_v1.0.kicad_sch

Title: Stepper Motor Evaluation Board

Size: A3	Date: 2026-10-23	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/5

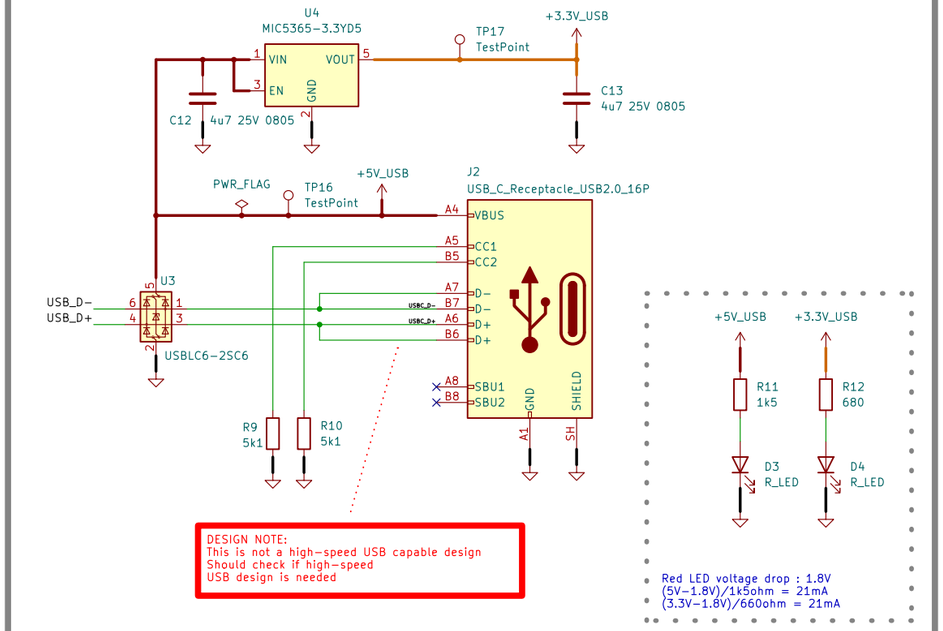
MCU

ELECTRICAL CHARACTERISTICS:
3V3 30 mA
-> should be checked with stm tool



USB

DESIGN NOTE: USB not isolated could be in final design



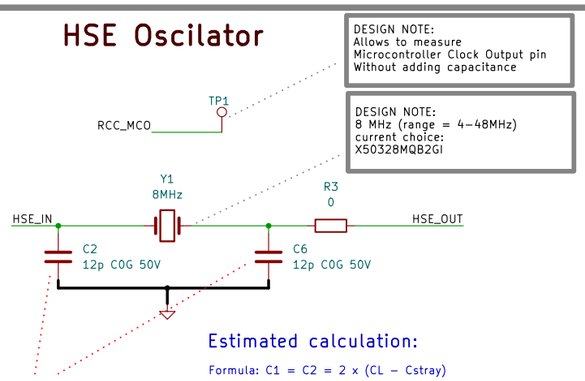
HSE Oscilator

DESIGN NOTE: No need of LSE oscilator since no sleep activity

DESIGN NOTE: HSE oscilator since space and BOM allows it
Better accuracy timing for everything including UART and USB

DESIGN NOTE: There are no constraints on crystal size, then it is recommended to choose a crystal with the smallest possible ESR.

DESIGN NOTE: If there are no constraints on power consumption, it is recommended to choose crystals with high load capacitance.



DESIGN NOTE: Allows to measure Microcontroller Clock Output pin Without adding capacitance

DESIGN NOTE: 8 MHz (range = 4-48MHz) current choice: X50328MQB2GI

DESIGN NOTE: Capacitors values should still be calculated depending on PCB traces etc.

Use 0402 footprint for easy replacement of caps when testing

Estimated calculation:

Formula: C1 = C2 = 2 x (CL - Cstray)

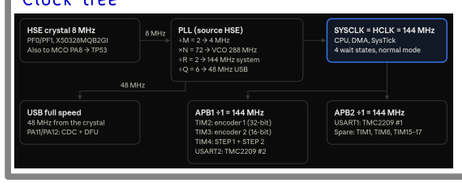
with Cstray estimated = 10pF p.100 datasheet MCU
C1 and C2 should be between 5pF-20pf

X50328MQB2GI:
CL = 16pF -> so that C1, C2 are in middle of allowed range
ESR = 80ohm
Tolerance = +-10ppm
Drive level = 100µW or higher

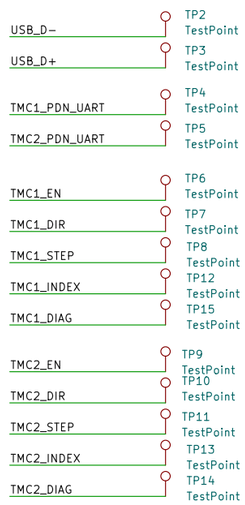
C1 = C2 = 12pF (estimate) -> good in middle of range

LAYOUT NOTE:
- place crystal and caps very close to OSC pins
- keep traces short and symmetrical
- avoid noisy digital routing near the oscillator loop
- avoid excessive vias in crystal traces
- respect the datasheet's ESR and drive recommendations

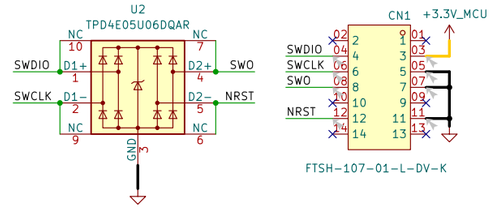
Clock tree



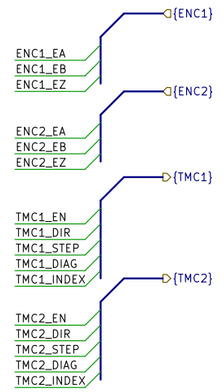
Testing points



SWD (Debug) Port



Interconnections



Sheet: /MCU - IO/
File: mcu.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/5

Stepper motor drivers

ELECTRICAL CHARACTERISTICS:
Driver current depending on motor: 1A/phase with 2ohm phase resistance
Pmotor_coil = (1A)² x (2ohm) = 2W
Motor has two coils so : 2W x 2 = 4W per motor
Idriver = P/U = 4W / 24V = 0.167A
Two drivers in total so : 0.167A x 2 = 0.33A

Two drivers = 24V 0.5A

One driver IO = 3V3 30µA
Two driver IO = 3V3 60µA

DESIGN NOTE:
Should still check layout example p82

DESIGN NOTE:
Do not power MCU output pin when IC not powered= when MCU only powered by USB

DESIGN NOTE:
Aluminium electrolytic capacitor

LAYOUT NOTE:
Place near IC with short path to die pad

DESIGN NOTE:
Chopper mode selection done via UART

DESIGN NOTE:
Diagnostic and StallGuard output.
Hi level upon stall detection or driver error.
Diagnostic done via UART

DESIGN NOTE:
In combination with a mechanical home switch, a more precise homing is enabled.
see p.8 and p66

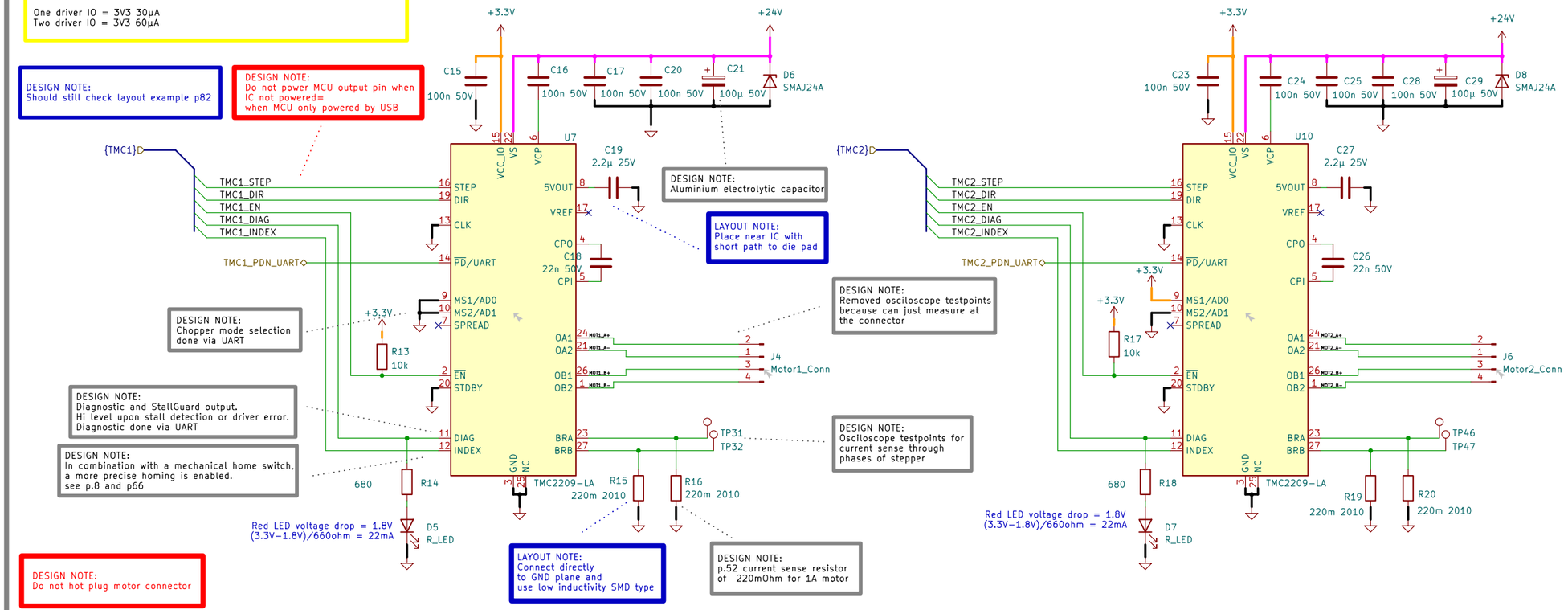
DESIGN NOTE:
Removed osciloscope testpoints because can just measure at the connector

DESIGN NOTE:
Osciloscope testpoints for current sense through phases of stepper

LAYOUT NOTE:
Connect directly to GND plane and use low inductivity SMD type

DESIGN NOTE:
p.52 current sense resistor of 220mOhm for 1A motor

DESIGN NOTE:
Do not hot plug motor connector



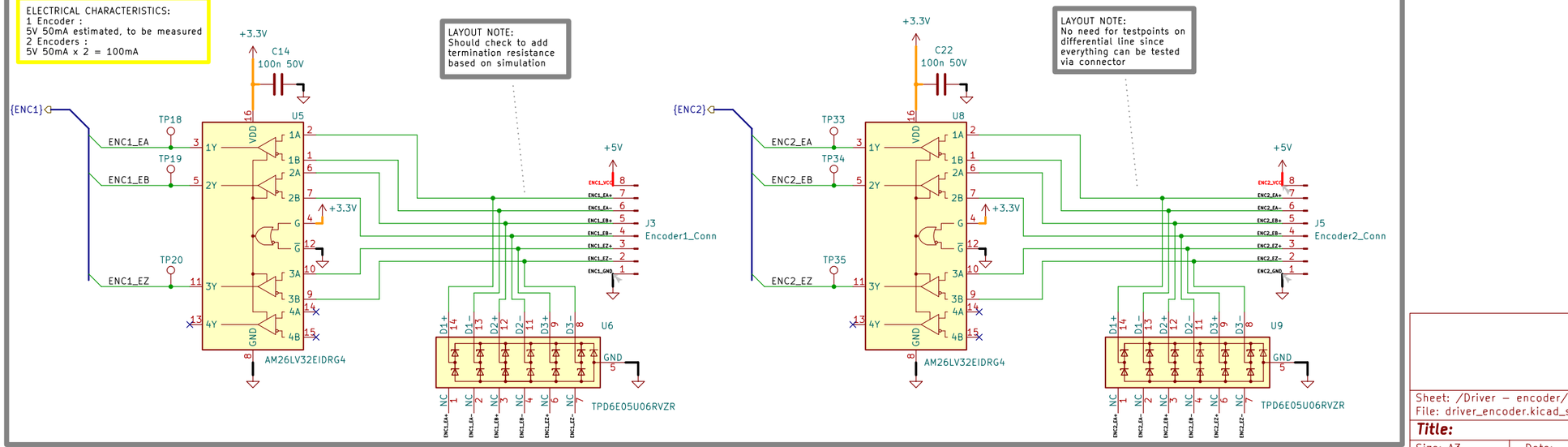
Encoder – RS422 differential signals to logic signals

ELECTRICAL CHARACTERISTICS
1 AM26LV32E :
3V3 17mA max supply current
2 AM26LV32E :
3V3 17mA x 2 = 34mA

ELECTRICAL CHARACTERISTICS:
1 Encoder :
5V 50mA estimated, to be measured
2 Encoders :
5V 50mA x 2 = 100mA

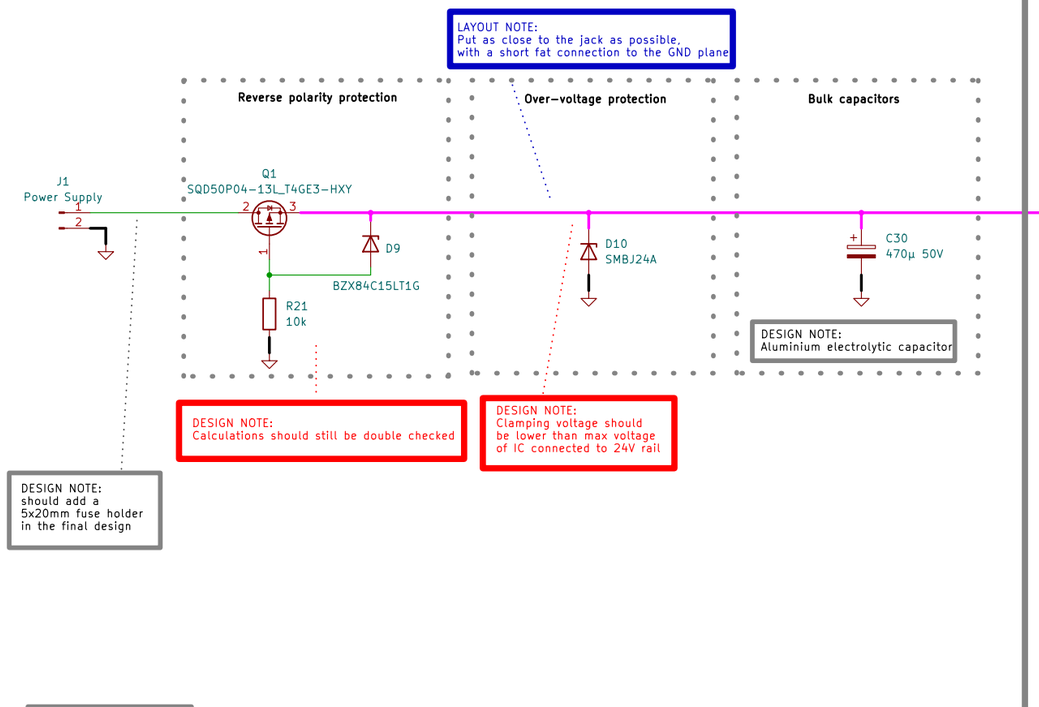
LAYOUT NOTE:
Should check to add termination resistance based on simulation

LAYOUT NOTE:
No need for testpoints on differential line since everything can be tested via connector

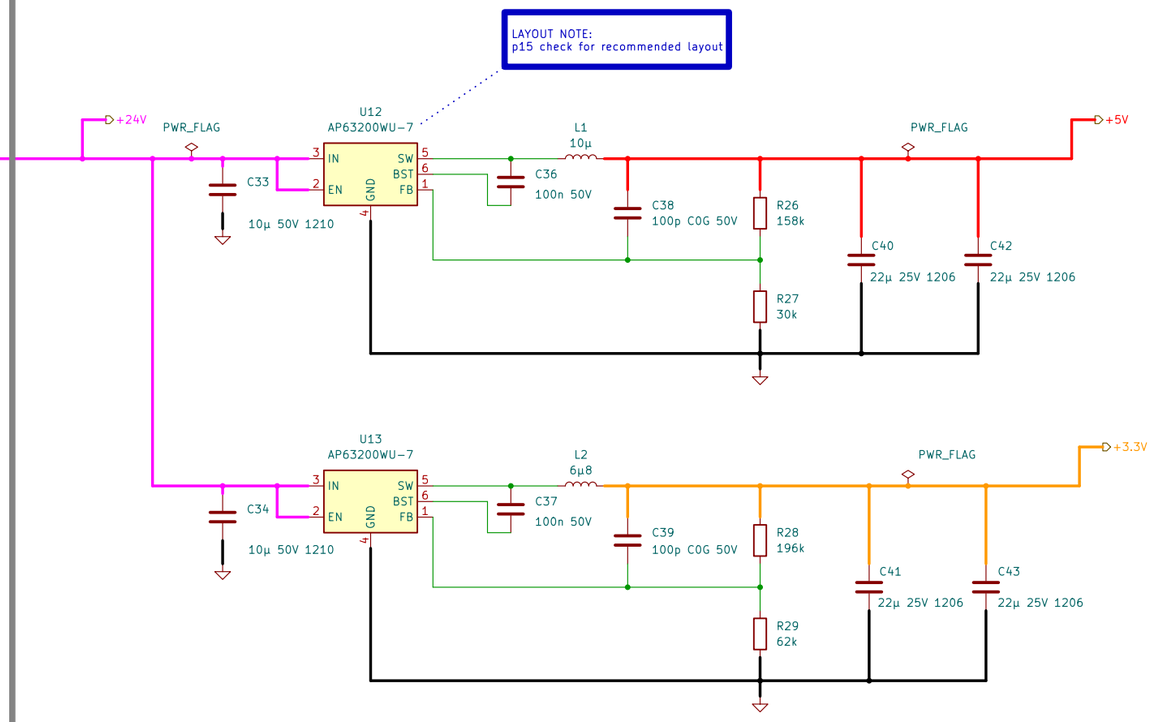


Sheet: /Driver – encoder/		
File: driver_encoder.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/5

Power supply protection circuitry



DC-DC Buck converters



Current consumption

Voltage rail	Components	Estimated current draw	Dimensioned current draw
24V	2x motor (driver)	0.5A	2A
5V	2x encoders	0.1A	1A
3V3	2x driver IC, RS422 signal converter	60µA + 34mA = 0.03406A	0.5A
5V_USB	MCU	30mA	0.5A

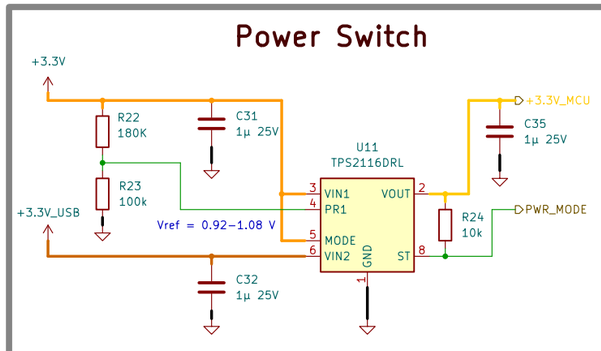
Component selector DC-DC buck converter

Table 1 shows a list of recommended component selections for common output voltages for the AP63200 and AP63201 referencing Figure 20.

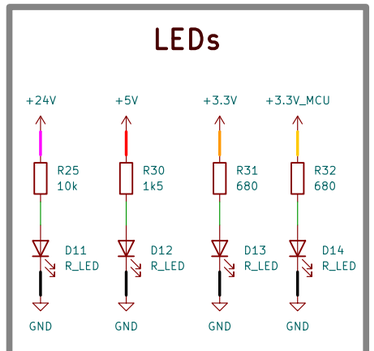
AP63200/AP63201							
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)	C4 (pF)
1.2	30.9	62	2.2	10	2 x 22	100	100
1.5	54.2	62	2.2	10	2 x 22	100	100
1.8	77.5	62	3.3	10	2 x 22	100	100
2.5	131	62	3.3	10	2 x 22	100	100
3.3	196	62	6.8	10	2 x 22	100	100
5	157	30	10	10	2 x 22	100	100
12	249	18	10	10	2 x 22	100	56

Table 1. Recommended Component Selections for AP63200/AP63201

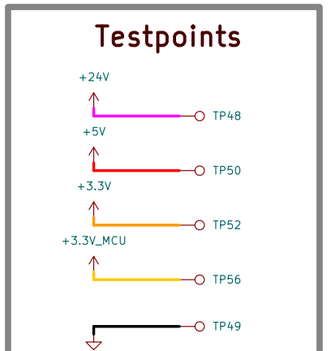
Power Switch



LEDs



Testpoints

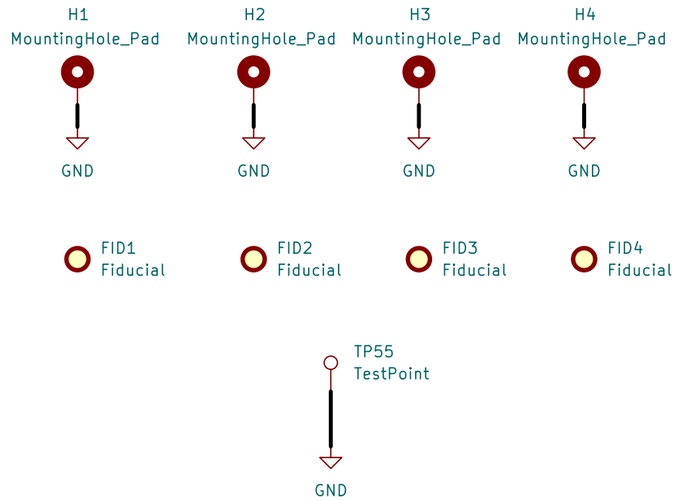


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Title:

Size: A3 | Date: | **Rev:**
KiCad E.D.A. 10.0.6 | Id: 4/5

Misc



Sheet: /Misc/
File: misc.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/5